

SECTION 13045-A CLEANROOM FILTERS

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. This Section specifies the requirements necessary to furnish and install:
 - 1. Cleanroom filter modules.
 - 2. Frames, mounting hardware, and accessories as indicated or specified.

1.2 RELATED WORK

- A. This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the specific requirements for cleanroom filters.
- B. CAUTION: Use of this Section without including all of the above-listed items will result in omission of basic requirements.
- C. In the event of conflict regarding requirements for cleanroom filters between this Section and any other section, the provisions of this Section shall govern.

1.3 CONSTRUCTION STANDARDS

- A. Work included in this Section shall conform to the following, except where specified otherwise within this Section:
 - 1. Federal Standard 209E.
 - 2. IES-RP-CC001.3.
 - 3. IES-RP-CC007.1.

1.4 DEFINITIONS

- A. Most Penetrating Particle Size (MPPS): The most penetrating particle size is that particular size that will most easily pass through the filter media. Hence, this is the particle size that will most easily pass through the filter media. The MPPS is a function of the filter medium structure, the air velocity, and particle composition.
- B. ULPA Filter: a scanned ultralow penetration air filter having a minimum particle collection efficiency of 99.9995 percent for the most penetrating particle size.

1.5 SYSTEM DESCRIPTION

- A. ULPA filter modules suitable for installation in cleanroom gel ceiling system as specified in Section 13042, Cleanroom Gel-Seal Ceiling System.
- B. The filter module shall consist of a continuous pleated filter media sheet, sealed in a filter frame and perforated face screen.

1.6 DESIGN CRITERIA

- A. Air Velocity: plus or minus 10 percent variance across the media face area at the rated airflow, measured 10 inches from the filter face. Uniformity shall be based on 12- by 12-inch readings over the entire filter face area.
- B. Dimensional Tolerance: plus or minus 1/16 inch on height, width, and length; plus or minus 1/8 inch on filter face diagonals.
- C. Pressure Drop: Variation in airflow resistance within 0.02 inch WC of each other.
- D. Access: filter modules shall be removable from the room side.
- E. Overcapacity: capable of operating at the rated efficiency with no structural degradation at twice the rated flow.
- F. Face Screen: minimum 50 percent free area and 0.05 inch WC maximum pressure drop.

1.7 QUALITY ASSURANCE

- A. The Owner and Contractor shall maintain the right to tour the manufacturer's facilities any time that fabrication is being performed on filters intended for this project.
- B. Filter modules shall be construction Grade 3 (UL-900 Class 1) as per IES-RP-CC001.3.
- C. Each filter shall have manufacturer's label so that all filter materials and tests can be tracked.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Factory clean, assemble, and package filter modules to comply with Section 01355, Clean Zone General Requirements.

- B. Deliver, store, and handle filter modules to comply with Section 01355, Clean Zone General Requirements, and Section 01600, Product Requirements.
- C. Always transport and store filters on their sides, never on the face or back sides
- D. Wrap filters in individual cartons, with protective sleeves inserted between the filter module and the carton walls to protect the bare media.
- E. Incorporate manufacturer's label on each carton indicating model number, serial number, media lot number, efficiency, rated capacity, total pressure drop, and test results for the individual filter module.
- F. Arrange cartons three across and three high, loaded on pallets, with hardboard sheet along both long sides of the pallet. Strap the entire pallet and cover with protective polyethylene wrap.
- G. Never stack filter pallets.
- H. Provide a readily visible shock sticker on each filter pallet.
- I. Ship filter pallets directly to jobsite by air-ride-enclosed vans.
- J. Receive, unload, inspect, store, uncrate, and install filter modules at the jobsite.

1.9 SUBMITTALS

- A. Provide the following with the Bid:
 - 1. Complete certified performance data at design conditions, including cfm and air pressure drop.
 - 2. Product literature and a schedule with quantities and sizes.
 - 3. Product installation details.
 - 4. Drawings showing ceiling grid interface detail.
 - 5. Product data listing chemical compositions of all construction materials.

1.10 EXTRA STOCK

- A. Provide an additional 5 percent of spare constructed filter modules to permit immediate replacement of damaged filters.

PART 2 - PRODUCTS

2.1 FILTER MODULE

- A. Media:
 - 1. Material: continuous pleated sheet of molded paper without seams, supports, or penetrations, self-supporting and rigid within the frame with no kinks.
 - 2. Separators: prohibited.
 - 3. Media paper thickness: 20-mil minimum.
 - 4. The filter pack shall be permanently sealed to the module frame.
- B. Face Screen: removable protective flush-mounted face screen with finish to match the ceiling grid.

2.2 SOURCE QUALITY CONTROL

- A. Factory Inspection:
 - 1. Visually inspect each filter for filter pleat straightness and uniformity.
 - 2. Measure each filter to verify dimensional tolerances.
 - 3. Test each filter for squareness by inserting a set square in each corner.
 - 4. Inspect each filter to verify that the manufacturer's label is attached and contains all required information.
 - 5. Reject filters that do not pass full factory inspection.
- B. Media Efficiency Test:
 - 1. Obtain three samples from each filter media slitter roll at the beginning, middle, and end.
 - 2. For ULPA filters, test media samples using a laser test; identify the MPPS.
 - 3. Reject rolls that do not achieve the rated efficiency on all samples.
- C. Filter Efficiency (Penetration) Test:
 - 1. Test each filter as described in IES-RP-CC007 for overall efficiency using a dual-laser particle counter with detection capability of 0.1 μm .
 - 2. Test filters at rated flow.

3. Sample upstream and downstream of filters simultaneously.
 4. Challenge Media: polystyrene latex spheres in the size range of the MPPS.
 5. Reject filters that fail to achieve the rated efficiency.
- D. Filter Pressure Test:
1. Test each filter for pressure drop at the rated airflow rate.
 2. Reject filters with pressure drops exceeding the rated performance requirements by 10 percent.
 3. Test ten randomly selected filters for performance and structural integrity at twice the rated flow. Mark these filters to distinguish them from other filters.
- E. Filter Leak (Scan) Test:
1. Test each filter at rated air flow for leaks using the particle counter scanning test method described in IES-RP-CC001.3, Section 9.2.2, 1 inch from filter face. Detection capability shall be 0.2 μm or less.
 2. Scan each filter with overlapping strokes to cover the entire filter area, with separate passes made around the entire periphery along the adhesive bond between the filter and frame.
 3. Challenge media: polystyrene latex spheres sized 0.2 μm and greater with a concentration of 30 by 10^7 particles per cubic foot.
 4. Sample flow rate: 1 cfm.
 5. Sample probe: size to maintain the sample flow rate and not exceed the maximum scan rate of 6 fpm, calculated using the formula in IES-RP-CC001.3.
 6. Spot-verify readings equal to or greater than half the maximum allowable leak rate. The spot verification shall be for 30 seconds. If the total particle count for the spot verification exceeds the maximum allowable leak rate, repair and retest the filter.
- F. Filter Velocity Uniformity Test:
1. Test 10 filters to confirm compliance with the design criteria for velocity uniformity.
 2. Measure velocity using a rotary anemometer or Velgrid.
 3. Reject filters that fail to meet the design criteria.

2.3 FINISHES

- A. Comply with Section 13042 - Cleanroom Gel-Seal Ceiling System.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Inspect pallets and shock stickers when received at the jobsite. Note defects and damage and recommend corrective action. Inspect and accept filters as specified in Section 01355, Clean Zone General Requirements.
- B. Uncrate filter modules only when ready for final inspection and installation.
- C. Reject pallets with broken shock stickers and remove from the jobsite.

3.2 INSTALLATION

- A. Install filters under clean construction protocol Level 4 as specified in Section 01355, Clean Zone General Requirements.
- B. Install in accordance with manufacturer's written instructions.

3.3 MATERIAL PREPARATION AND CLEANING

- A. Clean filter modules as specified in Section 13007, Clean Zone Cleaning Procedures.

3.4 FIELD QUALITY CONTROL

- A. Performance testing and certification shall be in accordance with Section 13009, Cleanroom Certification.

- END -